

Temperature-Controlled Transformation of WO₃ Nanowires into Active Facets-Exposed Hexagonal Prisms toward Efficient Visible-Light-Driven Water Oxidation

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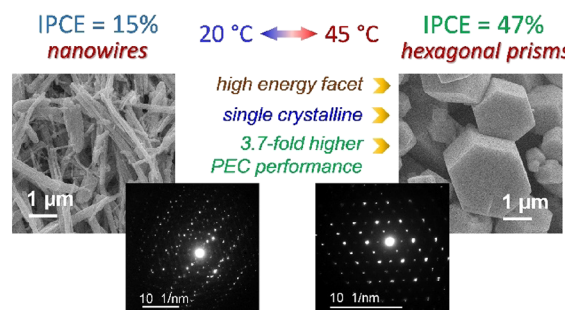
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ABSTRACT: A unique transformation of WO₃ nanowires (NW-WO₃) into hexagonal prisms (HP-WO₃) was demonstrated by tuning the temperature of the (N₂H₄)WO₃ precursor suspension prepared from tungstic acid and hydrazine as a structure-directing agent. The precursor preparation at 20 °C followed by calcination at 550 °C produced NW-WO₃ nanocrystals (*ca.* <100 nm width, 3–5 μm length) with anisotropic growth of monoclinic WO₃ crystals to (002) and (200) planes and a polycrystalline character with randomly oriented crystallites in the lateral face of nanowires. The precursor preparation at 45 °C followed by calcination at 550 °C produced HP-WO₃ nanocrystals (*ca.* 500–1000 nm diameter) with preferentially exposed (002) and (020) facets on the top-flat and side-rectangle surfaces, respectively, of hexagonal prismatic WO₃ nanocrystals with a single-crystalline character. The HP-WO₃ electrode exhibited the superior photoelectrochemical (PEC) performance for visible-light-driven water oxidation to that for the NW-WO₃ electrode; the incident photon-to-current conversion efficiency (IPCE) of 47% at 420 nm and 1.23 V vs RHE for HP-WO₃ was 3.1-fold higher than 15% for the NW-WO₃ electrode. PEC impedance data revealed that the bulk electron transport through the NW-WO₃ layer with the unidirectional nanowire structure is more efficient than that through the HP-WO₃ layer with the hexagonal prismatic structure. However, the water oxidation reaction at the surface for the HP-WO₃ electrode is more efficient than the NW-WO₃ electrode, contributing significantly to the superior PEC water oxidation performance observed for the HP-WO₃ electrode. The efficient water oxidation reaction at the surface for the HP-WO₃ electrode was explained by the high surface fraction of the active (002) facet with fewer grain boundaries and defects on the surface of HP-WO₃ to suppress the electron–hole recombination at the surface.

KEYWORDS: tungsten trioxide, photoanode, nanostructure, facet engineering, water splitting, photoelectrocatalysis

Temperature-controlled transformation of WO₃ photoanode



1. INTRODUCTION

The ever-growing energy crises and environmental issues associated with carbon emissions have compelled our society to develop efficient, clean, and renewable energy resources.^{1,2} Converting solar energy into hydrogen (H₂) fuel by photoelectrochemical (PEC) water splitting has attracted immense attention as an artificial photosynthesis technology to address those issues.^{3–8} In the PEC water splitting electrolyzers, the fuel generation efficiency is severely constrained by the sluggish kinetics of water oxidation at photoanodes.^{8–10} Over four decades, intensive research has been focused on n-type semiconductor photoanode materials with visible-light responses (band gap energy, $E_g \leq 2.8$ eV) such as Ta₃N₅,^{11–13} α-Fe₂O₃,^{9,14–16} BiVO₄,^{6,17–19} TaON,^{20–22} and WO₃.^{7,8,23–26} WO₃ photoanodes have been widely studied in PEC water oxidation owing to its appropriate band gap ($E_g \approx 2.6$ eV), satisfactory valance band edge position for water oxidation

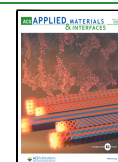
(~ 3.0 V vs RHE), good photocarrier mobility (~ 12 cm² V^{−1} s^{−1}), and attractive stability under acidic conditions under solar illumination.^{7,8,23–32} Nevertheless, WO₃ photoanodes usually suffer from obstruction associated with separation, migration, and reaction of the photon-generated holes during the water oxidation process.^{7,33}

Great efforts have been dedicated to improve the PEC water oxidation performance of WO₃ photoanodes, such as surface passivation,^{34–36} nanostructure engineering,^{7,8,24,27,37,38} crystal facet engineering,^{28,29,39,40} heterojunction construction,^{41–43}

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selective doping,^{44–49} and oxygen evolution cocatalyst loading.^{26,33,50} Nanostructure control of WO₃ photoanodes have been recognized as an effective way to (1) increase the active sites at the surface–electrolyte interface, (2) reduce the hole migration length required for surface reactions, and (3) improve the electron transport for highly efficient PEC water oxidation.^{23,24} Despite the unrelenting efforts on nanostructure control of WO₃, the growth orientation control and facet engineering of WO₃ nanocrystals has been lesser explored to improve PEC water oxidation performance.^{7,28,29,51} Actually, intrinsic activities of semiconductor photoanodes depend strongly on the configuration of surface atoms and bonding environment that can be changed by controlling crystal facets.^{7,28,29,47–49} By the combination of experiments and theory, recent studies have revealed that the (002) facets of WO₃ nanocrystals have higher intrinsic activities than the other facets owing to the highest oxygen atom density.^{28,29,51} WO₃ nanocrystals with exposed (002) crystal facets have been developed by proper selection of a structure directing agent and/or tuning of synthesis parameters.^{7,28,29}

Recently, we emphasized trailblazing works on nanostructure engineering of WO₃ to achieve novel organized mesoporous structure^{8,27} and N₂-intercalated nanorods^{37,38} of crystalline WO₃ photoanodes by the template-directed strategy. We synthesized WO₃ nanorod crystals with *in situ* intercalation of N₂ into the lattice of WO₃ using hydrazine (N₂H₄), which acts as a versatile structure-directing agent, and the WO₃ nanorod photoanode drastically improved the performance of PEC water oxidation compared to unspecified WO₃ structures.^{37,38} In this synthesis, N₂H₄ serves the dual functional role of a source of N atoms for the *in situ* N₂ intercalation as well as a structure-directing agent for the nanorod architecture. The N₂ intercalation was found to provide a considerable red shift of the light absorption edge of WO₃ due to a new intermediate N 2*p* orbital formed between the conduction band (CB) and valence band (VB) of WO₃. Nevertheless, the formation mechanism of WO₃ nanocrystals directed by N₂H₄ still have not been investigated, although N₂H₄ is an effective structure-directing agent to design efficient WO₃ photoanode by nanostructure control of WO₃ crystals. Herein, we report a unique transformation of WO₃ nanowires (NW) (*via* the nanorods) into hexagonal prisms (HP) with specific facet-oriented growth by tuning temperature of the precursor synthesis using N₂H₄. The hexagonal prismatic WO₃ nanocrystals with a preferentially exposed (002) and (020) facets on two different surfaces for the HP-WO₃ electrode significantly improve the performance of PEC water oxidation compared to the polycrystalline WO₃ nanowires with random growth of nanocrystals for the NW-WO₃ electrode.

2. EXPERIMENTAL SECTION

2.1. Materials. The indium tin oxide conducting glass sheet (ITO, sheet resistance: 10 Ω/sq) was purchased from Asahi Glass Co. Tungstic acid (H₂WO₄) (Kanto Chemical Co.), hydrazine monohydrate (N₂H₄·H₂O), polyethylene glycol (PEG; Mw = 20,000) (Wako Pure Chemical Co.), and Marpolose 60MP-50 (Matsumoto Yushi-Seiyaku Co.) were purchased. All reagents are of analytical grade and used as received. Millipore water (18 MΩ·cm) was used for all the experiments.

2.2. Synthesis of N₂H₄-Derived Precursors. N₂H₄-derived precursors were synthesized by reactions under controlled temperatures. Tungstic acid (300 mg, 1.2 mmol) was dispersed in water (1.5 mL) and stirred vigorously at various temperatures from 20 to 45 °C. To the yellow suspension, N₂H₄·H₂O (146 μL, 3.0 mmol) was

gradually added dropwise. The pH values of the suspension prepared at 20 and 45 °C were *ca.* 9.3. After continuous stirring for 30 min, the resulting white suspension was filtered and dried under vacuum to obtain the NW- and HP-(N₂H₄)WO₃ precursor powder.

2.3. Fabrication of NW- and HP-WO₃ Electrodes. A total of 100 mg of (N₂H₄)WO₃ powder, 100 mg of polyethylene glycol (PEG, Mw = 20,000; binder), and 10 mg of merpolose (viscosity enhancer) were mixed in 400 μL of methanol. After stirring the mixture for ~1 h, all air bubbles disappeared to give a smooth paste. The paste was coated by a squeegeeing technique on an ITO substrate with a fixed area of 1.0 cm² and thickness of 100 μm by an applicator (YOSHIMITSU SEIKI, YBA I-type) and dried in air for 15 min. The as-coated ITO substrate was calcined under flowing O₂ in a two-step heating protocol: first, the temperature was raised from 25 to 200 °C with a rate of 5.8 °C min^{−1} and then kept for 1 h; finally, the temperature was raised to 550 °C with a rate of 10 °C min^{−1} and then kept for 2 h to give the NW- and HP-WO₃ electrodes, respectively.

2.4. Characterization. Powder X-ray diffraction (XRD) were measured by a diffractometer (Rigaku, MiniFlex 600) using monochromated Cu Kα (λ = 1.54 Å) radiation. Diffraction data were recorded in the range of 2θ = 5–100° in 0.02° steps with a scan rate of 0.2° min^{−1}. Electron microscopy observation was carried out by a field-emission scanning electron microscope (FE-SEM; JEOL, JSM-6500F, operated at 10 kV and working distance of 10 mm) and a transmission electron microscope (TEM; JEOL, JEM-ARM200F, operated at 200 kV). To prepare the TEM specimen for HP-WO₃, the HP-WO₃ crystals were embedded in a resin attached to a Si substrate. The resultant composite was cut as a thin slice (several 100 nm) parallel to the flat-hexagonal surface of HP-WO₃ crystals by a focus ion beam (FIB) processing technique and placed on the TEM grid using a *in situ* sample manipulator (OmniProbe 400; Oxford instruments). Nitrogen sorption isotherms were measured using a surface area analyzer (BEL Japan Inc., BELSORP-miniII) at 77 K. The samples were degassed under vacuum for 3 h at 300 °C, prior to gas adsorption. The Brunauer–Emmett–Teller (BET) method was used to calculate the surface areas. X-ray photoelectron spectroscopy (XPS) was performed by an X-ray photoelectron spectrometer (Ulvac, Phi 5000 Versa Probe CU) using a monochromated Al Kα (1486.6 eV; 350 W) radiation. The emitted photoelectrons were collected with path energy of 23.5 eV through a hemispherical analyzer. During the measurements, the pressure was below 1 × 10^{−9} Torr in the analyzing chamber. The binding energies in XPS were calibrated using the C 1s peak appearing at 284.8 eV. UV–visible diffuse reflectance spectra (DRS) were obtained on a spectrophotometer (JASCO V-670) in a DR mode with an integrating sphere (ISN-723). Raman spectra were recorded using a Raman spectrophotometer (Horiba-Jobin-Yvon, LabRAM HR) using 532 nm excitation and silicon standard wavenumber (520.7 cm^{−1}).

2.5. PEC Measurement and Product Analysis. An argon atmosphere and 25 °C temperature were maintained to conduct all the photoelectrochemical measurements using an electrochemical analyzer (Hokuto Denko, HZ-7000). A standard three-electrode cell equipped with NW- or HP-WO₃ electrodes, Pt wire, and Ag/AgCl as working, counter, and reference electrodes, respectively, was used for the photocurrent measurements. An electrolyte solution of 1.0 M HClO₄ (pH = 0) was employed for all the measurements. The potentials (*E*_{RHE}) were expressed versus a reversible hydrogen electrode (RHE), converted from the measured potentials (*E*_{Ag/AgCl}) versus Ag/AgCl and the standard potential (*E*^o_{Ag/AgCl} = 0.199 V vs SHE at 25 °C) of Ag/AgCl according to eq 1:

$$E_{\text{RHE}} = E_{\text{Ag/AgCl}} + E^{\circ}_{\text{Ag/AgCl}} + 0.059 \text{ pH} \quad (1)$$

The potentials were expressed as the values versus RHE unless noted otherwise. All the photoelectrochemical measurements were performed under visible light (λ > 390 nm, 100 mW cm^{−2}) from a 500 W xenon lamp (Ushio Inc., Optical ModuleX) with a UV-cut filter (L39) and a liquid filter (0.2 M CuSO₄, 5.0 cm light pass length) cutting of heat ray. The output of light intensity was calibrated as 100 mW cm^{−2} using a spectroradiometer (Ushio Inc., USR-40). The

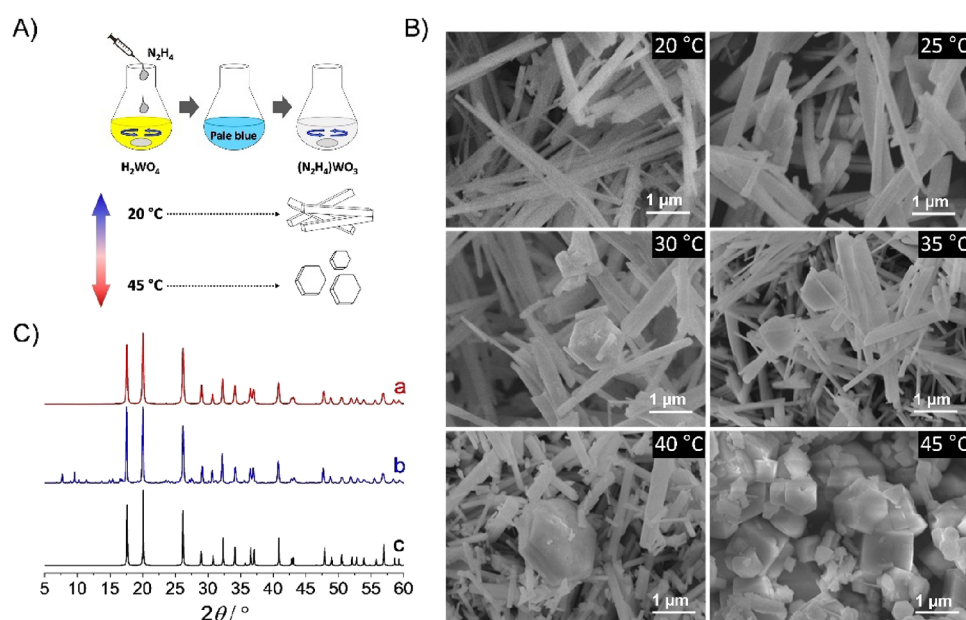


Figure 1. (A) Schematic representation for temperature-controlled synthesis of NW- and HP- $(\text{N}_2\text{H}_4)\text{WO}_3$ precursors. (B) SEM images of $(\text{N}_2\text{H}_4)\text{WO}_3$ precursors prepared at different controlled temperatures. (C) XRD patterns of $(\text{N}_2\text{H}_4)\text{WO}_3$ precursors prepared at (a) 20 and (b) 45 °C. (c) XRD pattern of trigonal- $(\text{N}_2\text{H}_4)\text{WO}_3$ derived from crystallographic information file (CIF) of CCDC number 1846977.⁵⁴

illuminated area was limited to 1.0 cm^2 , and the light was irradiated from the ITO side (backside illumination). The photocurrents were recorded either through the positive direction sweeping of the potential with a scan rate of 10 mV s^{-1} or through applying a constant potential. Mott–Schottky plots were obtained in a potential range of 0.2 to 0.8 V vs Ag/AgCl at a frequency of 0.5 kHz. Electrochemical impedance (EIS) measurement was carried out at an applied potential of 1.23 V vs RHE under the visible light in a frequency range of 100 mHz to 20 kHz (amplitude of 50 mV). IPCE measurement was performed at 1.23 V vs RHE in 1 M HClO_4 solution (pH = 0) in the three-electrode setup. A monochromatic light with 20 nm of bandwidth was given from a 500 W xenon lamp using a monochromator (M10, JASCO) for IPCE measurements. IPCE (%) values at each wavelength were calculated using eq 2:

$$\text{IPCE (\%)} = 1240 \times j \times 100 / P \times \lambda \quad (2)$$

where j , P , and λ refer to the photocurrent density (mA cm^{-2}), incident photon flux (mW cm^{-2}), and wavelength (nm), respectively. Bulk photoelectrolysis for water oxidation was performed in a gas tight two-compartment cell separated by a Nafion membrane. NW- or HP- WO_3 working and Ag/AgCl reference electrodes were placed in one compartment, and a Pt wire counter electrode was placed in the other compartment. In both compartments of the cell, 1.0 M HClO_4 (3.3 mL) was used as an electrolyte solution and it was sealed and saturated with Ar gas prior to the measurement. The gas phases (headspaces) of both the working and counter electrode compartments were analyzed for the evolved O_2 and H_2 , respectively, using gas chromatography (Shimadzu, GC-8A with a TCD detector and molecular sieve 5 Å column and Ar carrier gas). PEC oxidation of benzyl alcohol was performed in a two-compartment cell separated by a Nafion membrane, referring to the literature.⁵² NW- or HP- WO_3 working and Ag/AgCl reference electrodes were placed in one compartment, and a Pt wire counter electrode was placed in the other compartment. An electrolyte solution (12.0 mL) of 0.5 M Na_2CO_3 (adjusted to pH 3.0 by HNO_3) was used in both compartments of the cell, which were sealed and purged with Ar gas for 1 h prior to the measurement. 0.1 mmol benzyl alcohol was injected into the electrolyte solution in the working electrode compartment with continuous stirring. Bulk photoelectrolysis for benzyl alcohol was performed at 1.23 V vs RHE for 2 h, and the reaction solution was analyzed after different time intervals using gas chromatography–

mass spectrometry (GC–MS; Shimadzu, GCMS-QP2010 Ultra) with an Rtx-1 column (60 m $\times$ 0.32 mm inner diameter, 5 μm film thickness; Restek Corp., USA) and helium as carrier gas.

3. RESULTS AND DISCUSSION

3.1. Preparation and Characterization of N_2H_4 -Derived Precursors. The preparation is based on an acid–base reaction between tungstic acid (H_2WO_4) and N_2H_4 under the controlled temperature from 20 to 45 °C (Figure 1A), by which the yellow suspension of H_2WO_4 produced the white solid precursors, through a pale-blue intermediate attributed to the reduction of W^{6+} species in H_2WO_4 by N_2H_4 .⁵³ The morphologies of the precursors prepared at various temperatures have been observed by an FE-SEM technique (Figure 1B). The SEM image of the precursor obtained at 20 °C is composed of nanowires with several micrometers in length and <100 nm in width. With the increase of temperature from 20 to 30 °C, the nanowires were truncated and gradually transformed into rod-shaped structures. The hexagonal prism structure started to appear from 30 °C, and the specimen prepared at 45 °C is only composed of hexagonal prisms of 500–1000 nm in diameter. The XRD patterns (Figure 1C) of both the NW- and HP-shaped precursors prepared at 20 and 45 °C, respectively, exhibited a trigonal phase of crystalline $(\text{N}_2\text{H}_4)\text{WO}_3$ (CCDC no. 1846977), in which N_2H_4 moieties are located in voids between tetrahedral WO_4 units.⁵⁴ However, the XRD pattern of the HP- $(\text{N}_2\text{H}_4)\text{WO}_3$ precursors suggests that it is not a single compound but primarily consist of trigonal $(\text{N}_2\text{H}_4)\text{WO}_3$ as the peaks intensities of trigonal phase are similar to that of the single compound NW- $(\text{N}_2\text{H}_4)\text{WO}_3$. These results indicate that the temperature in the precursor preparation has significant impact on the growth orientation control of crystals of $(\text{N}_2\text{H}_4)\text{WO}_3$ precursors. The degree of specific interaction between the N_2H_4 moieties and the WO_4 tetrahedral units could vary with the temperature, which probably directs the growth orientation of the nanostructure of $(\text{N}_2\text{H}_4)\text{WO}_3$.

3.2. Preparation and Characterization of NW- and HP-WO₃ Electrodes. The NW- and HP-(N₂H₄)WO₃ precursors were coated on ITO electrodes by a squeegee technique with the aid of organic additives and then calcined at 550 °C under O₂ to give the WO₃ electrodes (identified by XRD patterns, *vide infra*), which are denoted as NW- and HP-WO₃ electrodes, respectively. The cross-sectional SEM images (Figure S1) showed no significant difference in the thickness of WO₃ coated layers for both the electrodes (average thickness estimated as *ca.* 7 μm). The top-view SEM images of the calcined electrodes revealed that the morphologies of the corresponding NW- and HP-(N₂H₄)WO₃ precursors remain intact over a domain of several micrometers without any noticeable deformation (Figure 2a,c). However, the higher

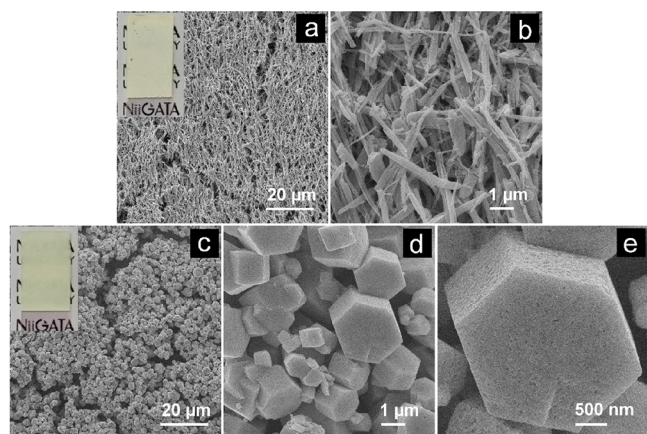


Figure 2. SEM images of (a, b) NW-WO₃ and (c, d, e) HP-WO₃ electrodes. The photographs of the corresponding electrodes are shown in the inset.

magnification SEM image of the NW-WO₃ specimen (Figure 2b) revealed that nanowires with a length of 3–5 μm were stacked after calcination. Meanwhile, the surface of the HP-WO₃ prominently appeared to be porous after calcination (Figure 2e), which perhaps attributed to the release of N₂/NH₃ gas during thermal decomposition of the HP-(N₂H₄)WO₃ precursor.⁵⁴ The N₂ sorption analysis (Figure S2) also showed the higher BET surface area (8.2 m² g⁻¹) of HP-WO₃ than that (3.6 m² g⁻¹) of NW-WO₃ owing to the porous surface of the HP nanostructure.

The XRD patterns (Figure 3A) revealed that both NW- and HP-WO₃ crystallized to phase-pure monoclinic WO₃ (JCPDF file no. 01-083-0950). The diffraction peaks at $2\theta = 23.1, 23.6,$ and 24.4° correspond to the (002), (020), and (200) planes with the characteristic *d*-spacings of 3.844, 3.769, and 3.648 Å, respectively. Very interestingly, the peaks corresponding to the (002) and (200) planes are higher than that to the (020) plane for NW-WO₃, being different from HP-WO₃, which shows nearly the same intensities of the three peaks, agreeing with the PDF data of monoclinic WO₃ (Figure. 3B). This indicates anisotropic crystallization of monoclinic WO₃ with predominant growth to (002) and (200) planes for NW-WO₃, in contrast to isotropic crystallization for HP-WO₃. For reference, WO₃ nanorods prepared similarly with N₂H₄ in our previous reports shows nearly same intensity ratio of the three peaks, indicating isotropic crystallization for WO₃ nanorods.^{37,38} This emphasizes the unique character of the anisotropic crystallization of monoclinic WO₃ for NW-WO₃. The higher peak intensity of the (002) reflection for NW-WO₃ compared with

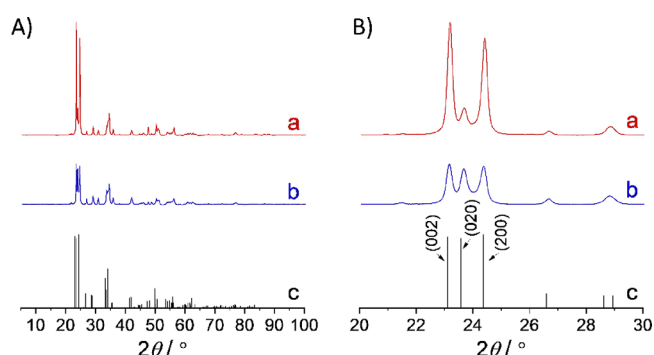


Figure 3. (A) XRD patterns of (a) NW- and (b) HP-WO₃ electrodes. (B) Magnified XRD patterns of corresponding samples. (c) JCPDF pattern of monoclinic WO₃ (file number: 01-083-0950) is shown by vertical drop-lines.

that for HP-WO₃ is attributed to the higher average crystallite diameter for NW-WO₃. The average crystallite diameters were estimated as 29.8 and 27.6 nm for NW- and HP-WO₃,⁵⁵ respectively from (002) reflections according to the Scherrer equation with a Bragg constant of 0.9. The lattice parameters *a*, *b*, *c*, and β of monoclinic crystal phases for NW- and HP-WO₃ were calculated by the whole-powder-pattern fitting (WPPF) method (Figure S3), and the results are summarized in Table S1. No substantial change was shown in the lattice parameters of *a*, *b*, *c*, and β for NW-WO₃ (*a* = 7.319437 Å, *b* = 7.536103 Å, *c* = 7.698277 Å, and β = 90.45649°) and HP-WO₃ (*a* = 7.3233(4) Å, *b* = 7.5284(3) Å, *c* = 7.6948(4) Å, and β = 90.487(4)°), which suggests that two different nanostructures and different growth of crystal planes for NW- and HP-WO₃ have not significantly affected their WO₃ lattice structure.

To further understand the crystal facets, transmission electron microscopy (TEM) analysis was conducted. The TEM images showed stacked nanowires (*ca.* 80–100 nm width; Figure 4a) and crystal of hexagonal prism (*ca.* 1000 nm width; Figure 4d) for NW- and HP-WO₃, respectively, which are in agreement with the respective SEM observations. The pores on the HP-WO₃ surface (Figure 4d) were clearly visible as bright spots due to cutting of the crystals as a thin slice for the preparation TEM specimen. The high-resolution transmission electron microscopy (HRTEM) image of the lateral face of nanowires for NW-WO₃ disclosed a polycrystalline character of crystallites (Figure 4b). A short-range-crystalline structure with several line and surface defects were clearly visible throughout the nanowire framework. The lattice fringes were measured to be 3.82 Å, which is close to the *d*-spacing value of the (002) plane of monoclinic WO₃. The selected area electron diffraction (SAED) patterns of NW-WO₃ revealed many discrete diffraction spots, signifying short-range ordering of crystallites in the polycrystalline WO₃ framework (Figure 4c). In a sharp contrast, the HRTEM image of HP-WO₃ viewed from the top of the flat-hexagonal-surface revealed an uninterrupted single-crystalline character without any grain boundaries (Figure 4e). The interplanar distances of the lattice fringes were observed to be 3.72 and 3.60 Å, corresponding to the *d*-spacing values of the (020) and (200) planes of monoclinic WO₃, respectively; however, the lattice fringes (*ca.* 3.8 Å) for the (002) plane cannot be observed. This suggests that the (002) facet most possibly exists on the flat-hexagonal-surface of the hexagonal prism because in a monoclinic WO₃ crystal structure, a (002) plane is oriented with an angle of *ca.* 90° to both (020) and (200) planes. The SAED patterns of

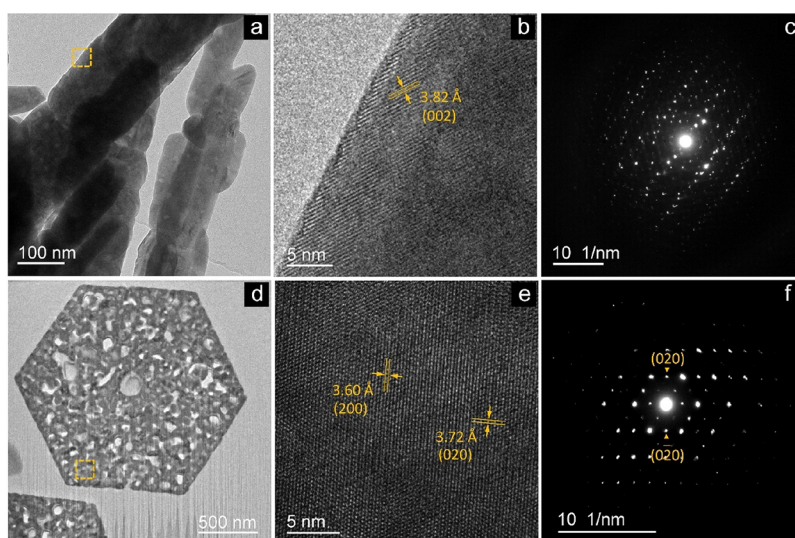


Figure 4. TEM and HRTEM images of (a, b) NW-WO₃ and (d, e) HP-WO₃. SAED pattern of (c) NW-WO₃ and (f) HP-WO₃. The images for HP-WO₃ were viewed from the top of the flat-hexagonal-surface.

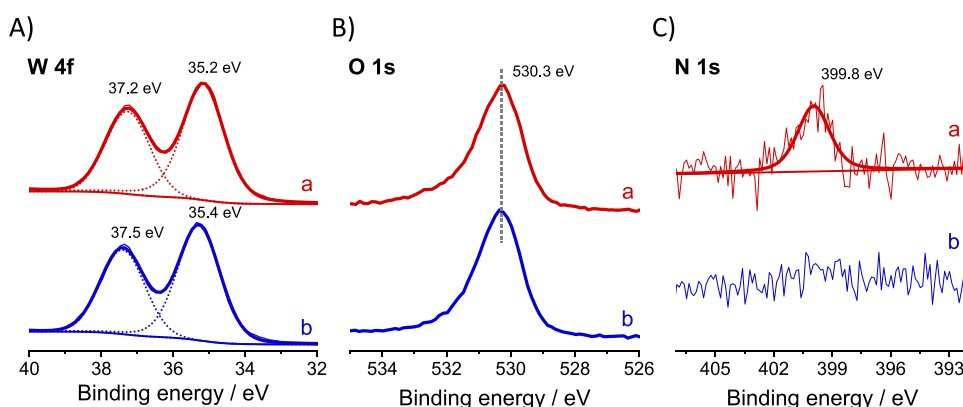


Figure 5. XPS spectra recorded in (A) W 4f, (B) O 1s, and (C) N 1s regions for (a) NW- and (b) HP-WO₃.

HP-WO₃ (Figure 4f) viewed from the top of the flat-hexagonal-surface showed a single-crystalline character with prominent diffraction spots arranged in a hexagonal pattern with the interplanar distance of 3.72 Å, corresponding to the (020) and (0 $\bar{2}$ 0) planes. The SAED patterns implied that the (020) facet is most likely exposed on the side-rectangle-surface of the hexagonal prism, which is consistent with the observation of the interplanar distances of the (020) (3.72 Å) and (200) planes (3.60 Å) on the flat-hexagonal-surface in the lattice fringe analysis. The hexagonal prism morphology was preformed in the trigonal (N₂H₄)WO₃ precursor, which is transformed into single-crystalline WO₃ with keeping a hexagonal arrangement of the monoclinic crystallites after calcination, as confirmed by the XRD and SEM analysis.

The X-ray photoelectron spectroscopic (XPS) measurement was conducted to reveal the chemical composition and valence state of these differently grown WO₃ nanocrystals. The XPS spectra in the W 4f region (Figure 5A) are almost identical between NW- and HP-WO₃ and deconvoluted to two peaks (37.2 and 35.2 eV for NW-WO₃; 37.5 and 35.4 eV for HP-WO₃), which are assigned to the binding energies of W 4f_{5/2} and W 4f_{7/2} in the WO₃ lattice, respectively.^{37,56} The XPS spectra in the O 1s region (Figure 5B) are also identical between NW- and HP-WO₃, exhibiting a peak at 530.3 eV, which was assigned to the binding energy of oxygen atoms in

the lattice.^{57,58} Moreover, the XPS spectrum in the N 1s region for NW-WO₃ exhibited a very weak peak at 399.8 eV, whereas a nearly negligible N 1s peak was detected for HP-WO₃ (Figure 5C). The peak at 399.8 eV for NW-WO₃ is assigned to the nitrogen bound at the surface, perhaps with oxygen atoms of WO₃,^{59,60} which is distinguishably different from the nitrogen bound with the tungsten center for intercalated N₂ (ca. 396.9 eV) as observed for WO₃ nanorods prepared similarly with N₂H₄ in our previous reports.^{37,38} The polycrystalline character of NW-WO₃ could offer more defects on the surface for the nitrogen bound on the surface. By contrast, the surface of HP-WO₃ could give lesser possibilities for the nitrogen bound on the surface for HP-WO₃ due to its single-crystalline character. The resonance Raman spectra depicted no evidently distinguishable Raman signals for different vibrational modes of the monoclinic WO₃ lattice of NW- and HP-WO₃ (Figure S4A and Table S2). Moreover, any detectable Raman signal was not observed in a range of 2200–2400 cm^{−1} (Figure S4B), which is different from the clearly observed Raman signals at 2327–2340 cm^{−1} for N≡N vibration of N₂ intercalated in the WO₃ lattice for the previously reported WO₃ nanorods.^{37,38} Thus, these XPS and resonance Raman data demonstrated that the chemical composition, the valence state, and the lattice environment of WO₃ nanocrystals were not significantly affected by their

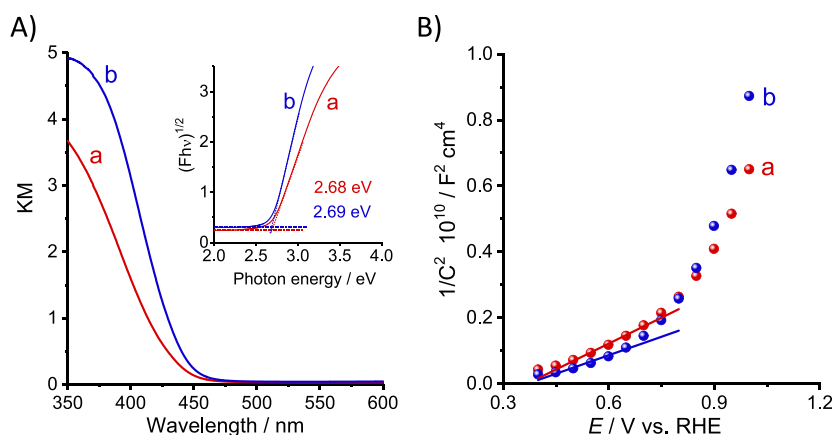


Figure 6. (A) UV–visible diffuse reflectance spectra (DRS) as a Kubelka–Munk (KM) function and (B) Mott–Schottky plots at 0.5 kHz in 1.0 M HClO₄ solution (pH = 0) of (a) NW- and (b) HP-WO₃ electrodes. Inset of (A) shows corresponding Tauc plots.

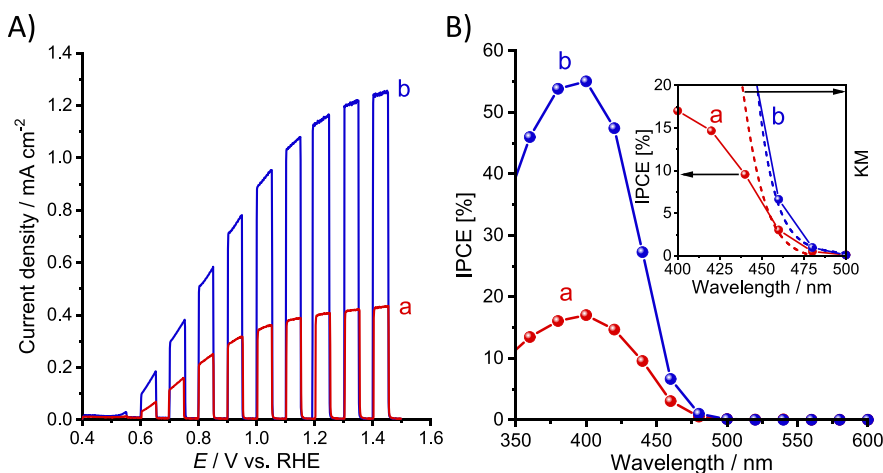


Figure 7. (A) Linear sweep voltammograms (LSVs) of (a) NW- and (b) HP-WO₃ electrodes as measured in 1.0 M HClO₄ solution (pH = 0) with chopped visible-light irradiation ($\lambda > 390 \text{ nm}$, 100 mW cm^{-2}) in a positive potential scan direction. (B) Action spectra of IPCE at 1.23 V vs RHE. Inset shows the demagnified IPCE plots along with corresponding DRS spectra.

different nanostructures and growth orientations of crystal facets for NW- and HP-WO₃, which is also in agreement with the XRD analysis of lattice parameters. These data also reveal that N₂ molecules are not intercalated in the WO₃ lattice of NW- and HP-WO₃, being different from the case of the previously reported WO₃ nanorods.^{37,38}

The UV–visible diffuse reflectance spectra (DRS) and the corresponding Tauc plots are shown in Figure 6A. The DRS of NW- and HP-WO₃ exhibited the comparable absorption edges at $\sim 470 \text{ nm}$. The band gap energies estimated from the Tauc plots (Figure 6A, inset) were 2.68 and 2.69 eV for NW- and HP-WO₃, respectively, which are in agreement with the band gap energy of most WO₃ reported previously.^{7,24,27,28,38} The Kubelka–Munk (KM) values in the range from 350 to 450 nm for HP-WO₃ are 1.3–1.6 times higher than those for NW-WO₃, showing the higher light absorption property of HP-WO₃. To investigate the semiconductor properties of NW- and HP-WO₃ electrodes, the capacitances ($C/F \text{ cm}^{-2}$) were measured with different applied potentials at a frequency of 0.5 kHz in 1.0 M HClO₄ solution (pH = 0). Mott–Schottky plots (plots of $1/C^2$ vs the applied potential; Figure 6B) depicted typical behaviors for n-type semiconductors, with linear increases of the reciprocal of the square of capacitance ($1/C^2 [F^{-2} \text{ cm}^4]$) with the applied potentials beyond the flat

band potentials. From the intercepts of the x axis and the slopes of the linear lines, the flat band potentials ($E_{\text{FB}}/V \text{ vs RHE}$) and the carrier densities (N_D/cm^{-3}) were calculated. The E_{FB} values and the N_D values of HP-WO₃ (0.34 V, $4.7 \times 10^{18} \text{ cm}^{-3}$) and NW-WO₃ (0.34 V, $3.4 \times 10^{18} \text{ cm}^{-3}$) are very close to each other and among the values ($E_{\text{FB}} = 0.32\text{--}0.39 \text{ V}$ and $N_D = \sim 1.0 \times 10^{18}\text{--}\sim 2.5 \times 10^{22} \text{ cm}^{-3}$) of most WO₃ electrodes reported previously.^{33,37,61–63} The very close band structure (in terms of band gap energies and E_{FB}) of two differently grown NW- and HP-WO₃ nanocrystals is consistent with their similar lattice environment.

3.3. PEC Water Oxidation Properties of NW- and HP-WO₃ Electrodes. The linear sweep voltammograms (LSVs) of NW- and HP-WO₃ electrodes with chopped visible light irradiation in a 1.0 M HClO₄ solution (pH = 0) are shown in Figure 7A. The photoanodic currents were observed above 0.60 V vs the reversible hydrogen electrode (RHE) for NW- and HP-WO₃ electrodes due to water oxidation, which was confirmed by O₂ evolution through gas chromatography (*vide infra*). However, the photocurrent at 1.23 V vs RHE for HP-WO₃ (1.16 mA cm^{-2}) was 2.8 times higher than that of NW-WO₃ (0.41 mA cm^{-2}). Figure 7B shows the action spectra of the incident photon-to-current conversion efficiency (IPCE) at 1.23 V vs RHE for NW- and HP-WO₃ electrodes. The onset

wavelength for generation of photocurrents were 480 nm for both electrodes, which is in agreement with band gap energies indicated from the DRS data of respective electrodes (Figure 6A). This shows that the photocurrent is generated by the band gap excitation of WO_3 . The IPCE values (IPCE_{420}) at 420 nm for HP- WO_3 (47%) was 3.1 times higher compared to that of NW- WO_3 (15%). The IPCE_{420} values were compared with those of the hitherto-reported WO_3 electrodes as summarized in Table S3.^{8,27,28,30,37,38,62,64–70} Although rigorous comparisons with the reported IPCE_{420} values are not easy because those were measured under different conditions such as pH of electrolyte solutions and applied potentials, the IPCE_{420} values for the HP- WO_3 (47%) electrode is noticeably higher or comparable with those of most hitherto-reported WO_3 photoanodes, except in the case of a WO_3 nanoplate electrode (65% at 1.0 V vs RHE in 1 M HClO_4).⁶⁸

The electrochemical surface area (ECSA) of NW- and HP- WO_3 electrodes was compared in Figure 8, which was derived

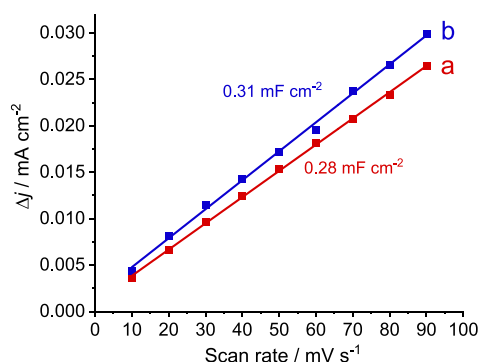


Figure 8. Plots of Δj versus scan rates for measurement of electrochemical surface area (ECSA) for (a) NW- and (b) HP- WO_3 electrodes. Δj is the difference between anodic (j_a) and cathodic (j_c) current densities at 0.84 V from CVs provided in Figure S5.

from the cyclic voltammograms (CVs in Figure S5) in a potential region of 0.8–0.9 V vs RHE (non-Faradaic region) under dark conditions. The scan rate dependence of the capacitive current in CVs (measured as the current density difference (Δj) between the anodic (j_a) and cathodic (j_c) ones at the potential of 0.84 V) provided the ECSA as the double-layer capacitance ($C_{dl}/\text{mF cm}^{-2}$) from the slopes of the linear plots of Δj versus scan rates (Figure 8).⁷¹ The C_{dl} value for HP- WO_3 (0.31 mF cm^{-2}) is higher by a factor of 1.1 than that of NW- WO_3 (0.28 mF cm^{-2}), which is in agreement with the higher BET surface area of the HP- WO_3 nanostructure. Nevertheless, the photocurrent density at 1.23 V vs RHE normalized by ECSA for HP- WO_3 is still 2.6 times higher than that of the NW- WO_3 electrode. The higher photocurrent generation for the HP- WO_3 electrode for PEC water oxidation cannot be explained by the difference in the interface area between the electrode surface and the electrolyte solution.

To reveal the origin of the superior performance of HP- WO_3 to NW- WO_3 for PEC water oxidation, photoelectrochemical impedance spectroscopic (PEIS) measurements were carried out. Nyquist plots of NW- and HP- WO_3 electrodes displayed semicircles in a frequency region of 100 mHz to 20 kHz, as shown in Figure 9. However, the semicircle of the HP- WO_3 electrode was significantly small compared to the NW- WO_3 electrode. Nyquist plots were analyzed by considering an equivalent circuit of two component sets including charge

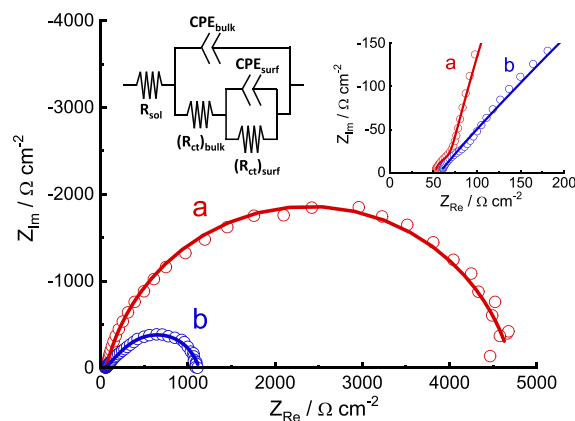


Figure 9. Nyquist plots for (a) NW- and (b) HP- WO_3 electrodes in 1.0 M HClO_4 solution (pH = 0) at 1.23 V vs RHE with visible light irradiation ($\lambda > 390$ nm, 100 mW cm^{-2}). Inset (right) shows the magnified Nyquist plots. Z_{Re} and Z_{Im} represent real and imaginary resistances, respectively. The solid lines show the simulated spectra by fitting to the impedance data using the equivalent circuit (inset, left). R_{ct} , R_{sol} , and CPE are the charge transfer resistance, solution phase resistance, and constant phase element, respectively. The subscripts of "bulk" and "surf" represent the components for the bulk charge transport through the film and the water oxidation reaction at the surface, respectively.

transfer resistance (R_{ct}) and constant phase element (CPE), in addition to solution phase resistance (R_{sol}) (Figure 9, inset). The two component sets were differentiated from each other by the subscripts "bulk" and "surf", assumed as components for the bulk charge transport through the film ($(R_{ct})_{bulk}$ and CPE_{bulk}) and the water oxidation reaction at the surface ($(R_{ct})_{surf}$ and CPE_{surf}), respectively.

The given parameters of the elements of the equivalent circuits are summarized in Table 1. The R_{sol} values were in a range of 52.6–59.0 $\Omega \text{ cm}^{-2}$, including the resistances of the electrolyte solution and the ITO electrode resistance. The $(R_{ct})_{bulk}$ value for the NW- WO_3 electrode (26.3 $\Omega \text{ cm}^{-2}$) was lower than that for the HP- WO_3 electrode (107 $\Omega \text{ cm}^{-2}$) by factors of 4.1 (Table 1). Concomitantly, the CPE constant ($(T_{CPE})_{bulk} = 1.9 \mu\text{F s}^{(p-1)}$, $p = 1.0$, p of a CPE exponent of non-ideality) of the CPE_{bulk} component for the NW- WO_3 electrode was lower than that ($(T_{CPE})_{bulk} = 34.4 \mu\text{F s}^{(p-1)}$, $p = 0.74$) for the HP- WO_3 electrode. This result suggests more efficient bulk electron transport through the NW- WO_3 layer with the unidirectional nanowire structure than that through HP- WO_3 layer with the hexagonal prismatic structure. However, the $(R_{ct})_{surf}$ value (989 $\Omega \text{ cm}^{-2}$) with $(T_{CPE})_{surf}$ of $49.5 \mu\text{F s}^{(p-1)}$ ($p = 0.76$) in the surface component for the HP- WO_3 electrode was 4.7 times lower than that (4680 $\Omega \text{ cm}^{-2}$) with $(T_{CPE})_{surf}$ of $21.4 \mu\text{F s}^{(p-1)}$ ($p = 0.84$) for the NW- WO_3 electrode (Table 1). This suggests more efficient water oxidation reaction at the surface for the HP- WO_3 electrode than the NW- WO_3 electrode, contributing significantly to the superior PEC water oxidation performance observed for the HP- WO_3 electrode. This is explainable by the crystal facets existed on the surface of HP- WO_3 nanocrystals. Recently, the computational studies demonstrated that the surface energies of (002) and (020) facets (1.56 and 1.54 J m^{-2} , respectively) are higher than that of the (200) facet (1.43 J m^{-2}) due to the higher oxygen atom density on (002) and (020) facets as compared to (200) facet of a monoclinic WO_3 nanocrystal, which reasonably explained the high intrinsic activities at (002)

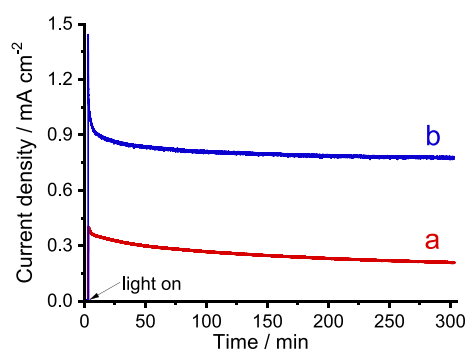
Table 1. Parameters of the Elements of R_s , R_{ct} , and CPE Analyzed in the Equivalent Circuit (Inset of Figure 9) for NW- and HP-WO₃ Electrodes^a

samples	R_s [Ω/cm^2]	surface			bulk		
		$(R_{ct})_{\text{surf}}$ [Ω/cm^2]	CPE_{surf}		$(R_{ct})_{\text{bulk}}$ [Ω/cm^2]	CPE_{bulk}	
			p	$(T_{\text{CPE}})_{\text{surf}}$ [$\mu\text{F s}^{(p-1)}$]		p	$(T_{\text{CPE}})_{\text{bulk}}$ [$\mu\text{F s}^{(p-1)}$]
NW-WO ₃	52.6	4680	0.84	21.4	26.3	1.0	1.9
HP-WO ₃	59.0	989	0.76	49.5	107	0.74	34.4

^aImpedance data were taken at 1.23 V vs RHE in a 1.0 M HClO₄ solution (pH = 0) with visible light irradiation ($\lambda > 390$ nm, 100 mW cm⁻²). p is a CPE exponent of non-ideality. T_{CPE} is a CPE constant.

and (020) facets for PEC water oxidation.^{29,51} It was also reported that the energy barriers for the OH dissociation steps of the water oxidation reaction at the surface are sufficiently low on the (002) facet than that on the (200) facet for a plate-shaped WO₃ nanocrystal.²⁸ The HRTEM image analysis in Figure 4 displays that crystal planes of NW-WO₃ are randomly oriented due to the polycrystalline growth of nanocrystals, which reduces the fraction of the active (002) facet on the nanowire surface. However, for HP-WO₃, the (002) and (020) facets were revealed to be preferentially exposed on the top-flat and side-rectangle surfaces of hexagonal prismatic nanocrystals, respectively. The high surface fraction of the active (002) and (020) facets is responsible for the more efficient water oxidation reaction at the surface for the HP-WO₃ electrode. Furthermore, the uninterrupted crystal structure of (002) and (020) facets with fewer grain boundaries and defects on the surface of HP-WO₃ (Figure 4d,e) could contribute to the suppressed electron–hole recombination to increase the observed water oxidation reaction at the surface.

The bulk photoelectrolysis for water oxidation was examined under potentiostatic conditions at 1.23 V vs RHE with visible light irradiation ($\lambda > 390$ nm) for 5 h. The current density–time profiles during photoelectrolysis using NW- and HP-WO₃ electrodes are displayed in Figure 10. After an initial spike in

**Figure 10.** Current density–time profiles during photoelectrolysis for water oxidation at 1.23 V vs RHE in 1.0 M HClO₄ solution (pH = 0) with continuous visible-light irradiation ($\lambda > 390$ nm, 100 mW cm⁻²) using (a) NW- and (b) HP-WO₃ electrodes.

the photocurrent density (associated to accumulation of holes and the capacitance component at the electrode–liquid interface), the photoanodic currents owing to water oxidation were observed for both NW- and HP-WO₃ electrodes. The HP-WO₃ electrode provided the distinctly superior anodic current to the NW-WO₃ electrode, which is corroborated by the high charge amount passed (14.6 C) and the amount (n_{O_2}) of O₂ evolved (26.4 μmol , 70% Faradaic efficiency (FE_{O_2})). The n_{O_2} value for HP-WO₃ was 3.7-fold higher than that of the NW-WO₃ (7.1 μmol , 59% FE_{O_2} ; charged amount 4.6 C). The

insufficient FE_{O_2} values (59 and 70%, respectively) for NW- and HP-WO₃ electrodes are possibly due to the competed photo-oxidation of the WO₃ surface to form tungsten-peroxo adducts,³³ commonly observed for WO₃-based photoanodes.^{8,30,33,72} The photo-oxidation of the WO₃ surface attenuates the photoanodic currents for NW- and HP-WO₃ electrodes, as illustrated in Figure 10. The photoanodic current density decreased at the beginning (20 min) of the photoelectrolysis for both NW- and HP-WO₃ electrodes (20% and 17%, respectively). However, the HP-WO₃ electrode was noticeably stable after the initial 20 min with ca. 90% of the current density (0.87 to 0.78 mA cm⁻²) retained from 20 min to 5 h, in contrast to ca. 66% of the current density (0.32 to 0.21 mA cm⁻²) retained in the same time range for the NW-WO₃ electrode. The photo-oxidation of the WO₃ surface could be attenuated at the active (002) and (020) crystal facets due to the suppressed hole accumulation by efficient water oxidation at the surface for the HP-WO₃ electrode, leading to the more stable photoelectrolysis for the HP-WO₃ electrode (especially after 20 min). Both NW- and HP-WO₃ electrodes were characterized before and after the bulk photoelectrolysis for 5 h using the SEM, XRD, and XPS techniques. No noticeable change in these experimental data was observed before and after the bulk photoelectrolysis (Figures S6–S8). The tungsten-peroxo adducts formed at photo-oxidation of the WO₃ surface cannot be detected in our XPS instrument.

Recently, the reactive oxidizing species such as H₂O₂, OH $\cdot$, etc. were demonstrated to be involved as key intermediates for PEC water oxidation on WO₃ photoanodes under acid conditions, which were utilized for the PEC oxidation of benzyl alcohol, methane, etc.^{52,73} To provide insight into the current stability in the bulk photoelectrolysis for PEC water oxidation (Figure 10), the catalytic performances for PEC oxidation of benzyl alcohol to benzaldehyde was investigated over NW- and HP-WO₃ electrodes. LSVs of NW- and HP-WO₃ electrodes in the presence and absence of benzyl alcohol (0.1 mmol) in a 0.5 M Na₂CO₃ solution (pH 3.0) with chopped visible light irradiation are shown in Figure 11A,B, respectively. In the presence of benzyl alcohol, the photoanodic current densities for the NW- and HP-WO₃ electrodes increased by a factor of 1.6 and 1.7 at 1.23 V vs RHE, respectively, with lowered onset shifts to 0.43 V for both electrodes, compared with the case in the absence of benzyl alcohol. However, the photocurrent densities at 1.23 V vs RHE for HP-WO₃ (3.0 mA cm⁻²) was 3.0 times higher than that of NW-WO₃ (1.0 mA cm⁻²) in the presence of benzyl alcohol, which is similar to the enhancement of the photocurrent for PEC water oxidation. Bulk photoelectrolysis for benzyl alcohol oxidation was examined under the potentiostatic conditions at 1.23 V vs RHE for 2 h using both electrodes. The current density–time profiles showed the significantly stable photo-

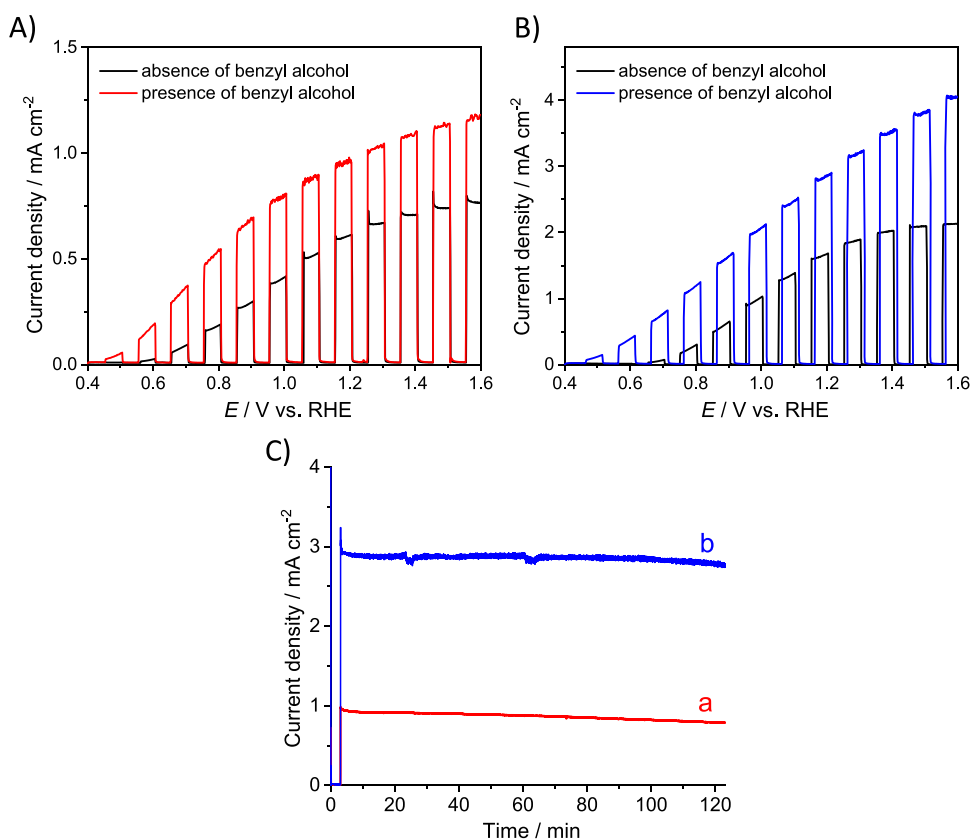


Figure 11. LSVs of (A) NW-WO₃ and (B) HP-WO₃ electrodes in the presence and absence of benzyl alcohol (0.1 mmol) in 0.5 M Na₂CO₃ solution (pH 3.0) with chopped visible-light irradiation ($\lambda > 390$ nm, 100 mW cm⁻²) in a positive potential scan direction. (C) Current density–time profiles during photoelectrolyses for benzyl alcohol oxidation at 1.23 V vs RHE in a 0.5 M Na₂CO₃ solution (pH 3.0) with the continuous visible-light irradiation using (a) NW- and (b) HP-WO₃ electrodes.

current during the 2 h photoelectrolysis for both NW- (0.95–0.79 mA cm⁻²) and HP-WO₃ electrodes (2.91–2.80 mA cm⁻²) (Figure 11C), in contrast to the current density decreased during the bulk photoelectrolysis for water oxidation, both NW- and HP-WO₃ electrodes (20% and 17%, respectively in initial 20 min, Figure 10). The product of the benzyl alcohol oxidation was detected as benzaldehyde by gas GC–MS without any detectable byproducts for both electrodes (Figure S9A,B). The amount of benzaldehyde produced gradually increased concomitantly with the decreased amount of residual benzyl alcohol (95–98% selectivity) and reached 0.74 mmol (67% yield based on initial benzyl alcohol of 0.11 mmol) after 2 h over HP-WO₃. This amount is *ca.* 2.9-fold higher than that (0.23 mmol, 23% yield at 2 h) over NW-WO₃ (Figure S9C). The Faradaic efficiencies (FE_{BA}) for benzaldehyde formation were calculated to be 72 and 81% for NW- and HP-WO₃ electrodes from the initial slopes (for 1 h) between the measured amounts of benzaldehyde and the theoretical amounts calculated from the charges required during the bulk photoelectrolyses. The deficient FE_{BA} (28 and 19%, respectively) for NW- and HP-WO₃ electrodes are attributed to water oxidation competed with the benzyl alcohol oxidation. The results on PEC oxidation of benzyl alcohol support that the decreased photoanodic current for PEC water oxidation is ascribed to photo-oxidation of the WO₃ surface by the reactive oxidizing species such as H₂O₂ and OH• to form tungsten-peroxo adducts.³³

4. CONCLUSIONS

A unique transformation of NW-WO₃ into HP-WO₃ was demonstrated by tuning the temperature of the (N₂H₄)WO₃ precursor suspension prepared from tungstic acid and hydrazine as a structure-directing agent. Nanowire-shaped and hexagonal prismatic WO₃ nanocrystals were attained by calcination at 550 °C from (N₂H₄)WO₃ precursors prepared at 20 and 45 °C, respectively. Nanowires for NW-WO₃ exhibited the anisotropic growth of monoclinic WO₃ crystals to (002) and (200) planes and unveiled a common polycrystalline character with randomly grown nanocrystals in the lateral face of nanowires, whereas hexagonal prisms for HP-WO₃ exhibited specific facet-oriented growth with (002) and (020) facets preferentially exposed on the top-flat and side-rectangle surfaces of the nanocrystals, respectively. The HP-WO₃ electrode gave an IPCE of 47% at 420 nm and 1.23 V vs RHE based on water oxidation, which is *ca.* 3.1 times higher compared to 15% for the NW-WO₃ electrodes. The high PEC water oxidation performance for the HP-WO₃ electrode is originated from the efficient water oxidation reaction at the surface on the highly intrinsically active (002) and (020) facets on the hexagonal prismatic nanocrystals. The present temperature-controlled nanostructure provides a facile way for facet engineering to develop efficient WO₃-based photoanodes and expected to be applied for a PEC water-splitting cell as artificial photosynthesis to improve the solar energy conversion efficiency.

■ ASSOCIATED CONTENT

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding authors.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.2c22483>.

Table for comparison, SEM images, N₂ sorption isotherms, XRD patterns and fitting analysis, resonance Raman spectra, XPS spectra, CVs, and GCMS profiles (PDF)

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Notes

The authors declare no competing financial interest.

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